

LUSAKA, Zambia

WMO: 676650

Lat: 15.333S

Lon: 28.450E

Elev: 3779

StdP: 12.80

Time Zone: 2.00 (E02)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	44.9	46.6	29.8	27.4	71.3	33.7	32.5	68.3	20.4	66.7	18.6	67.3	3.3	120	0.314

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10	26.6	93.4	64.9	91.5	65.4	89.5	65.3	76.0	79.6	74.7	78.8	73.7	78.2	8.1	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
75.3	152.9	77.8	73.7	144.7	76.6	72.8	140.2	76.0	42.8	79.8	41.3	79.9	40.2	77.6	82.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.3	16.2	14.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	69.9	73.1	73.1	72.0	69.5	65.0	61.7	61.0	65.2	71.6	76.8	76.0	74.0
	DBStd	6.08	1.94	2.06	2.51	3.21	2.75	3.31	3.22	3.78	3.95	3.78	3.74	2.72
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	325	0	0	0	3	36	109	129	44	4	0	0	0
	CDD50	7260	717	646	683	585	464	350	341	471	649	830	781	743
	CDD65	2110	252	226	218	138	35	9	5	50	203	365	331	278
	CDH74	16889	1086	1000	1115	866	480	257	248	907	2339	4196	2825	1570
	CDH80	5722	206	168	211	139	38	14	19	193	961	2147	1188	438
(14) Wind	WSAvg	5.8	4.3	4.8	5.2	6.3	6.3	5.9	6.5	6.6	6.6	6.7	5.8	4.6
(15) Precipitation	PrecAvg	31.2	8.5	6.1	3.5	1.0	0.1	0.0	0.0	0.0	0.0	0.5	3.3	7.5
	PrecMax	41.4	17.0	13.1	8.1	6.0	1.2	0.1	0.0	0.0	0.4	2.2	8.3	14.0
	PrecMin	17.6	3.4	2.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	1.4	2.7
	PrecStd	6.9	3.2	2.8	2.3	1.5	0.3	0.0	0.0	0.0	0.1	0.6	1.4	3.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	87.5	87.6	88.2	85.9	83.7	82.3	82.5	87.7	93.2	97.9	94.7	90.0
		MCWB	67.8	68.7	66.5	67.2	63.7	61.6	61.8	62.5	61.8	63.9	66.1	66.3
	2%	DB	84.4	84.1	84.5	83.4	81.0	79.2	79.4	84.5	91.0	95.0	92.0	87.2
		MCWB	70.9	71.0	70.6	69.2	64.5	61.0	59.4	60.8	63.5	64.7	66.4	67.9
	5%	DB	82.5	82.0	82.5	81.8	79.1	77.2	77.1	82.3	88.2	92.9	89.8	84.4
		MCWB	71.1	71.9	71.2	69.1	64.6	60.6	58.6	60.4	63.0	65.2	66.7	69.8
	10%	DB	80.5	80.2	80.4	79.6	77.0	75.0	74.7	79.7	86.2	90.6	87.4	82.1
		MCWB	70.8	71.4	70.3	68.2	63.7	59.4	57.6	59.9	62.2	65.3	66.7	69.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.2	78.7	77.3	75.1	71.9	67.4	66.9	69.1	71.3	72.0	75.0	75.7
		MCDB	81.1	81.4	81.1	80.2	78.3	76.2	79.2	81.2	88.1	87.2	79.9	80.3
	2%	WB	75.1	76.4	74.7	73.0	68.9	65.4	62.6	64.6	67.6	69.7	73.4	74.7
		MCDB	79.1	78.7	79.3	78.6	76.3	74.5	75.1	79.1	84.4	87.3	78.6	79.4
	5%	WB	74.0	74.8	73.2	71.1	66.8	63.4	60.5	62.9	65.4	68.3	72.1	73.4
		MCDB	78.1	77.3	78.7	77.9	75.3	72.7	73.4	77.8	83.4	86.1	78.4	78.3
	10%	WB	72.8	73.3	71.7	69.5	64.9	61.2	59.0	61.1	63.6	67.2	70.8	71.8
		MCDB	77.0	76.4	76.8	76.4	73.9	71.2	71.3	76.4	82.2	85.0	78.5	76.4
(35) Mean Daily Temperature Range	5% DB	MDBR	15.1	15.2	17.3	19.5	24.6	26.2	26.8	28.0	28.0	26.6	20.1	16.1
		MCDBR	19.3	18.6	20.4	22.2	26.8	28.7	30.8	31.5	30.7	29.6	25.2	20.5
		MCWBR	8.8	9.2	10.7	10.4	13.6	13.5	14.8	13.0	12.7	11.7	10.1	8.7
	5% WB	MCDBR	15.1	14.9	17.0	19.7	24.6	25.6	27.5	28.0	28.4	26.7	18.4	15.7
		MCWBR	9.6	10.8	11.6	11.8	15.4	15.0	15.8	14.9	14.3	12.4	9.6	9.0
(40) Clear-Sky Solar Irradiance	taub		0.376	0.376	0.357	0.314	0.295	0.294	0.317	0.372	0.452	0.438	0.404	0.381
	taud		2.458	2.471	2.524	2.621	2.619	2.566	2.473	2.284	2.085	2.159	2.309	2.434
	Ebn at Noon		307	304	304	306	302	297	292	283	271	283	296	305
	Edh at Noon		38	37	34	30	28	28	31	41	53	50	44	39
(44) All-Sky Solar Radiation	RadAvg		1707	1738	1728	1724	1688	1538	1591	1849	2049	2141	1958	1756
	RadStd		163	185	146	132	75	77	91	53	62	111	111	141

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page